

Chapter 1. Sequential Circuits

Last Week

Excitation tables of flip-flops

Analysis of sequential circuits

Implementation of sequential circuits from given state tables

This Week

Steps of sequential circuit design

Steps of Sequential Circuit Design

1. Decision of how many states we need or which values should be set in the memory components according to the verbal definition.
2. Creating of state table and state diagram in order to model the system's behavior.
3. Decision of which type of flip-flop to use (if not specified) and creation of excitation tables.
4. Deriving logic equations from output and excitation tables and implementing the circuit.

Steps of Sequential Circuit Design

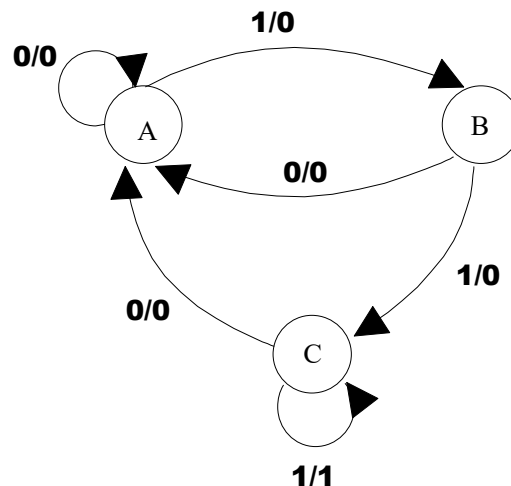
Example: Assume a Mealy type circuit with 1 input and 1 output. This circuit is expected to output 1 if and only if the last 3 input is 1.

This system has 3 states and these states are stored in memory components.

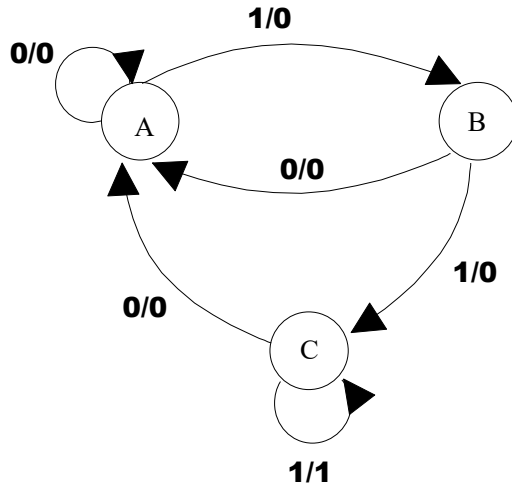
State A represents the state which last input is 0.

State B represents the state which last 2 inputs are 01 sequentially.

State C represents the state which last 2 inputs are 11 sequentially.



Example (2nd Page)



State Diagram

Present State (q)	Next State (Q)		Output (z)	
	x=0	x=1	x=0	x=1
A	A	B	0	0
B	A	C	0	0
C	A	C	0	1

State table of the circuit

Since the state table has 3 states, we need to use 2 flip-flops.

Let's assign letters as follows:

A = 00

B = 01

C = 10

Present State q_1q_2	Next State (Q_1Q_2)		Output (z)	
	x=0	x=1	x=0	x=1
0 0	0 0	0 1	0	0
0 1	0 0	1 0	0	0
1 0	0 0	1 0	0	1

Example (3rd Page)

Let's implement the circuit with JK type flip-flops.

Next step is to derive the inputs of flip-flops.

Present State q ₁ q ₂	Next State (Q ₁ Q ₂) x=0 x=1	Inputs of Flip-Flops				Output (z) x=0 x=1	
		x=0 J ₁ K ₁ J ₂ K ₂		x=1 J ₁ K ₁ J ₂ K ₂			
0 0	0 0 0 1	0 x	0 x	0 x	1 x	0	0
0 1	0 0 1 0	0 x	x 1	1 x	x 1	0	0
1 0	0 0 1 0	x 1	0 x	x 0	0 x	0	1



q Q	J K
0 0	0 x
0 1	1 x
1 0	x 1
1 1	x 0

Example (4th Page)

Let's draw a Karnaugh map for each input of flip-flops (J_1 , K_1 , J_2 , K_2) and the output (z) to find out the equations.

Present State q_1q_2	Next State (Q_1Q_2) $x=0$ $x=1$	Inputs of Flip-Flops				Output (z) $x=0$ $x=1$	
		$x=0$		$x=1$			
		J_1K_1	J_2K_2	J_1K_1	J_2K_2		
0 0	0 0 0 1	0 x	0 x	0 x	1 x	0	0
0 1	0 0 1 0	0 x	x 1	1 x	x 1	0	0
1 0	0 0 1 0	x 1	0 x	x 0	0 x	0	1

q_1q_2 x \	00	01	11	10
0			x	x
1		1	x	x

$$J_1 = x \cdot q_2$$

q_1q_2 x \	00	01	11	10
0		x	x	
1	1	x	x	

$$J_2 = x \cdot q_1'$$

q_1q_2 x \	00	01	11	10
0	x	x	x	1
1	x	x	x	

$$K_1 = x'$$

q_1q_2 x \	00	01	11	10
0	x	1	x	x
1	x	1	x	x

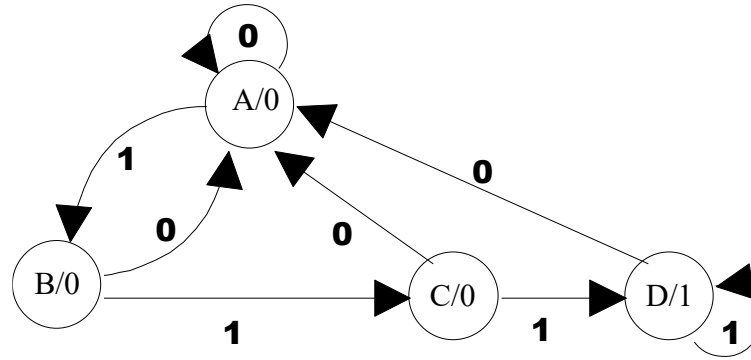
$$K_2 = 1$$

q_1q_2 x \	00	01	11	10
0			x	
1			x	1

$$z = x \cdot q_1$$

Steps of Sequential Circuit Design

Example: Assume a Moore type circuit with 1 input and 1 output. This circuit is expected to output 1 if and only if last 3 inputs are 1.

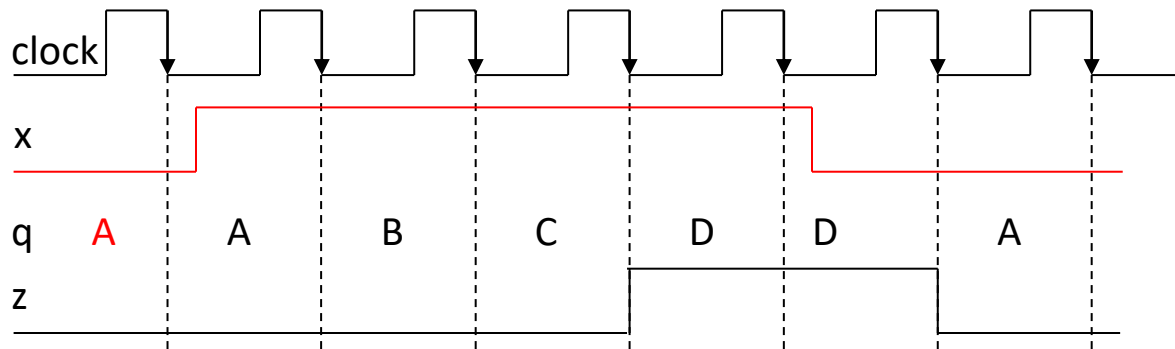


Assuming that the system's initial state is A, let's analyze the state transitions and the output.

A	A	B	C	D	D	D	D	A	B	C	A	B	C	D	A	B			
x	0	1	1	1	1	1	1	1	0	1	1	0	1	1	1	0	1		
z	0	0	0	0	1	1	1	1	1	0	0	0	0	0	0	1	0	0	0

Example (2nd Page)

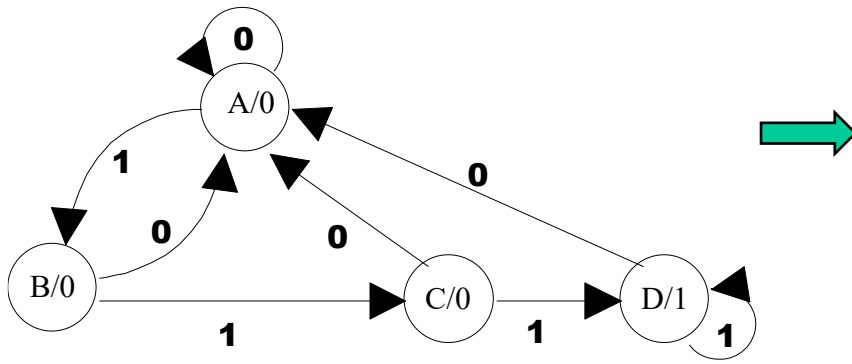
Let's analyze the behavior of the circuit on the time diagram.



Example (3rd Page)

And now, let's assign letters to flip-flops' outputs and derive the state table.

Let's assume that A=00, B=01, C=10, D=11.



Present State x q ₁ q ₂	Next State Q ₁ Q ₂	Inputs of Flip-Flops		Output z
		J ₁ K ₁	J ₂ K ₂	
0 0 0	0 0	0 x	0 x	0
0 0 1	0 0	0 x	x 1	0
0 1 0	0 0	x 1	0 x	0
0 1 1	0 0	x 1	x 1	1
1 0 0	0 1	0 x	1 x	0
1 0 1	1 0	1 x	x 1	0
1 1 0	1 1	x 0	1 x	0
1 1 1	1 1	x 0	x 0	1

Example (4th Page)

After drawing Karnaugh maps for J1, K1, J2, K2, and z, we find out the equations for each of them.

Present State $x \ q_1 \ q_2$	Next State $Q_1 \ Q_2$	Inputs of Flip-Flops		Output z
		$J_1 \ K_1$	$J_2 \ K_2$	
0 0 0	0 0	0 x	0 x	0
0 0 1	0 0	0 x	x 1	0
0 1 0	0 0	x 1	0 x	0
0 1 1	0 0	x 1	x 1	1
1 0 0	0 1	0 x	1 x	0
1 0 1	1 0	1 x	x 1	0
1 1 0	1 1	x 0	1 x	0
1 1 1	1 1	x 0	x 0	1

Karnaugh


$$\begin{aligned}
 J_1 &= q_2 \cdot x \\
 K_1 &= x' \\
 J_2 &= x \\
 K_2 &= x' + q_1 \\
 z &= q_1 \cdot q_2
 \end{aligned}$$

Since it is a Moore type circuit, the output only depends on the states.
 The output is 1 only in state D and we assigned D = 11.
 So, $z = q_1 \cdot q_2$.

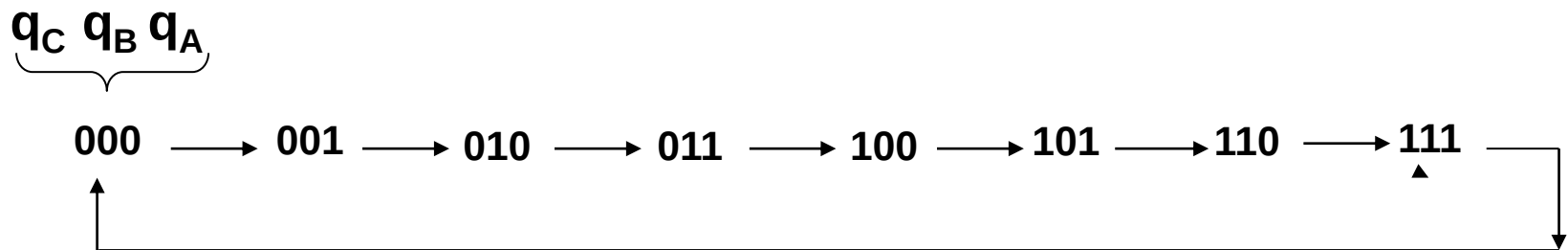
Steps of Sequential Circuit Design

Example: Synchronous Counter Design

Generally, counter circuits don't have inputs. State changes when clock pulse is received. The outputs are the states of flip-flops. In other words, flip-flops' outputs are also the circuit's outputs. If we need to control whether the counter will count up or down, we can add an input for it.

Design steps of a 3-bits binary counter:

Since this counter will count from 0 to 7, we'll need 8 states and therefore 3 flip-flops. Let's name these flip-flops as q_C (MSB), q_B and q_A (LSB).



Example (2nd Page)

Present State $q_C q_B q_A$	Next State $Q_C Q_B Q_A$	Inputs of Flip-Flops				
		$J_C K_C$	$J_B K_B$	$J_A K_A$		
0 0 0	0 0 1	0 x	0 x	1 x		
0 0 1	0 1 0	0 x	1 x	x 1		
0 1 0	0 1 1	0 x	x 0	1 x		
0 1 1	1 0 0	1 x	x 1	x 1		
1 0 0	1 0 1	x 0	0 x	1 x		
1 0 1	1 1 0	x 0	1 x	x 1		
1 1 0	1 1 1	x 0	x 0	1 x		
1 1 1	0 0 0	x 1	x 1	x 1		

With the help of Karnaugh maps, we find out that;

$$J_C = K_C = q_B \cdot q_A$$

$$J_B = K_B = q_A$$

$$J_A = K_A = 1$$



It's obvious that if we were to build a 4 bit counter, the equation of J_D would be as follows:

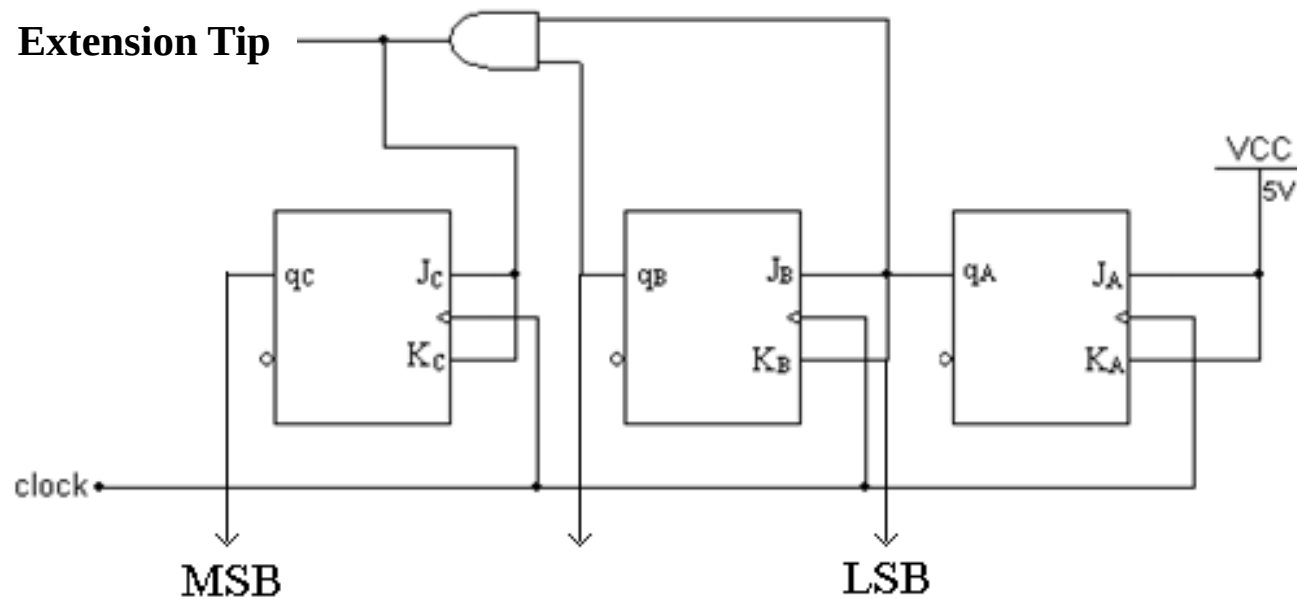
$$J_D = K_D = q_C \cdot q_B \cdot q_A$$

Example (3rd Page)

$$J_C = K_C = q_B \cdot q_A$$

$$J_B = K_B = q_A$$

$$J_A = K_A = 1$$



Circuit Diagram of a 3-bit Counter

Some Notes

1. In an up counter, if we use q' instead of q as circuit's output, then the circuit would act as a down counter, and vice versa.
2. If it isn't an ordered counter, then some states should be marked as don't care. But since a flip-flop's initial state is unknown, the initial state of the circuit might be one of the don't care states. This is an unwanted situation. To prevent such issues, flip-flops have Preset and Clear inputs in addition to D/T/SR/JK inputs. Preset and Clear inputs set and reset the output regardless of the clock signal, and are used to set initial state of the flip-flops.

